

# Linear SVM

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# Outline

Optimal Separating Hyperplane

Linear SVM

Lagrange Duality

Karush-Kuhn-Tucker conditions

# References

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- ▶ Trevor Hastie, Robert Tibshirani, Jerome Friedman (2009), "The Elements of Statistical Learning (ESL)", Chapter 4.4, 4.5, 12.2, 12.3
- ▶ S. Boyd and L. Vandenberghe (2004), "Convex optimization", Chapter 5.1, 5.2, 5.5
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# Optimal Separating Hyperplane

**Optimal Separating Hyperplane**

# Optimal Separating Hyperplane

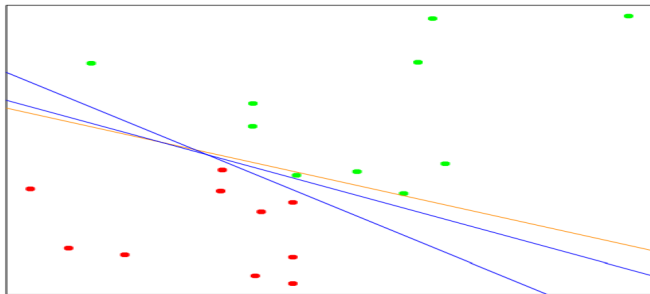
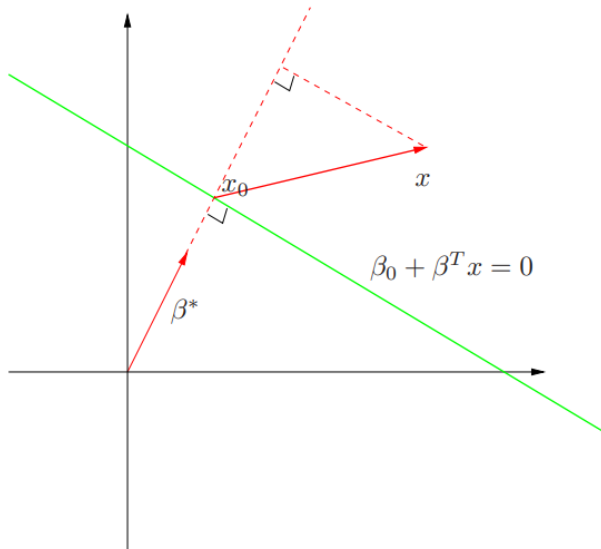


Figure 1: Figure 4.14 in ESLII

Suppose that we can separate the samples according to their labels without error using a linear decision function. How do we define the "best" linear decision function in this problem? Let's maximize the distance between samples and the separating hyperplane.

- ▶ How do we formulate this problem?
- ▶ Does this problem have a unique solution?

## Optimal Separating Hyperplane



## Optimal Separating Hyperplane

Given a hyperplane:

$$\{x \in \mathbb{R}^p | \beta_0 + \beta'x = 0\}$$

The distance from a point to a hyperplane is given by

$$\left\| \frac{\beta\beta'}{\|\beta\|^2}(x - x_0) \right\|$$

where  $x_0$  is any vector on the hyperplane. Note that  $\frac{\beta\beta'}{\|\beta\|^2}$  is the projection to the span of  $\beta$ . We have:

$$\begin{aligned} \left\| \frac{\beta\beta'}{\|\beta\|^2}(x - x_0) \right\| &= \frac{\|\beta\|}{\|\beta\|^2} |\beta'(x - x_0)| = \frac{1}{\|\beta\|} |\beta'(x - x_0)| \\ &= \frac{|\beta_0 + \beta'x|}{\|\beta\|} \end{aligned}$$

due to  $\beta_0 + \beta'x_0 = 0$ .

# Optimal Separating Hyperplane

Suppose we have the samples

$$x_i \in \mathbb{R}^p, y_i \in \{-1, 1\}, i = 1, \dots, n$$

The optimal separating hyperplane is found by maximizing the distance between the hyperplane and the samples.

$$\max_{\beta, \beta_0} M \quad \text{s.t.} \quad \frac{|x'_i \beta + \beta_0|}{\|\beta\|} \geq M, i = 1, \dots, n$$

where  $x' \beta + \beta_0 = 0$  is the separating hyperplane. WLOG, let's label the  $y_i$ 's so that the samples with  $y_i = 1$  lie on the upper side of the hyperplane. Then, we have

$$\max_{\beta, \beta_0} M \quad \text{s.t.} \quad \frac{y_i(x'_i \beta + \beta_0)}{\|\beta\|} \geq M, i = 1, \dots, n$$



## Optimal Separating Hyperplane

Since  $(\beta, \beta_0)$  and  $(c\beta, c\beta_0)$  yields a same decision boundary for  $c \neq 0$ , we set the restriction:  $\frac{1}{\|\beta\|} = M$

$$\max_{\beta, \beta_0} M \quad \text{s.t.} \quad y_i(x'_i\beta + \beta_0) \geq 1$$

Here, maximizing  $M$  is same as minimizing  $\frac{1}{2}\|\beta\|^2$ :

$$\min_{\beta, \beta_0} \frac{1}{2}\|\beta\|^2 \quad \text{s.t.} \quad y_i(x'_i\beta + \beta_0) \geq 1, \quad i = 1, \dots, n$$

- ▶ Samples with  $y_i(x'_i\beta + \beta_0) > 1$  are outside of the margin.
- ▶ Samples with  $y_i(x'_i\beta + \beta_0) = 1$  are right on the margin.

The linear function  $f(x) = \beta_0 + x'\beta$  defines the decision function. For prediction, we classify any  $x \in \mathbb{R}^p$  with  $x'\beta + \beta_0 \geq 0$  to coupled with  $y = +1$ , and  $x'\beta + \beta_0 < 0$  with  $y = -1$ ,

## Linear SVM

## Linear SVM

What if the samples are not linearly separable? Linear SVM is estimated by maximizing the distance between the hyperplane and samples using a soft margin.

$$\max_{\beta, \beta_0, (\xi_i)} M \quad \text{s.t.} \quad \begin{cases} \frac{y_i(x'_i\beta + \beta_0)}{\|\beta\|} \geq M(1 - \xi_i) \\ \xi_i \geq 0, \sum_{i=1}^n \xi_i \leq K \end{cases}$$

for a given  $K > 0$ . Here,

$$\frac{y_i(x'_i\beta + \beta_0)}{\|\beta\|}$$

is "signed" Euclidean distance, which will be negative if the sign of  $y_i$  disagrees with that of  $x'_i\beta + \beta_0$ .

- ▶  $\xi_i > 1$  means that the sample is on the wrong side of the hyper-plane.
- ▶  $1 \geq \xi_i > 0$  means that the sample is on the wrong side of the margin.
- ▶  $\xi_i = 0$  means that the sample is either correct side of the margin or right on the margin.

## Linear SVM

The value  $\xi_i$  is proportional to the amount the sample  $x_i$  is on the wrong side of the margin. When  $\xi_i > 1$ , the sample is on the wrong side of the decision function and misclassification occurs. Thus, bounding  $\sum_{i=1}^n \xi_i < K$  bounds the total number of training misclassifications at  $K$ .

# Linear SVM

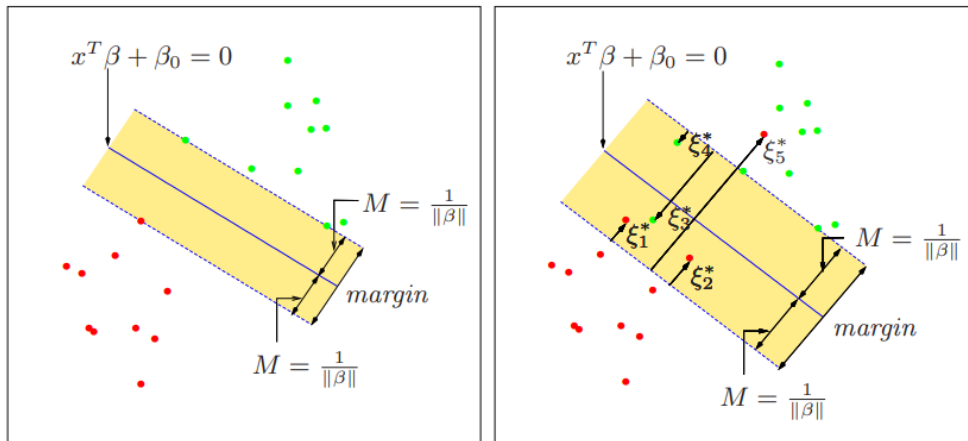


Figure 3: Figure 12.1 in ESLII

## Linear SVM

Again, since any  $\beta, \beta_0$  satisfying the constraint can be positively scaled without violating the constraint, we set the restriction:  $\frac{1}{\|\beta\|} = M$

$$\max_{\beta, \beta_0, (\xi_i)} M \quad \text{s.t.} \quad \begin{cases} y_i(x'_i\beta + \beta_0) \geq 1 - \xi_i \\ \xi_i \geq 0, \sum_{i=1}^n \xi_i \leq K \end{cases}$$

Since maximizing  $M$  is same as minimizing  $\frac{1}{2}\|\beta\|^2$ ,

$$\min_{\beta, \beta_0, (\xi_i)} \frac{1}{2}\|\beta\|^2 \quad \text{s.t.} \quad \begin{cases} y_i(x'_i\beta + \beta_0) \geq 1 - \xi_i \\ \xi_i \geq 0, \sum_{i=1}^n \xi_i \leq K \end{cases}$$

## Linear SVM

We can write this in the equivalent form:

$$\min_{\beta, \beta_0, (\xi_i)} \frac{1}{2} \|\beta\|^2 + C \sum_{i=1}^n \xi_i \quad \text{s.t.} \quad \begin{cases} y_i(x'_i \beta + \beta_0) \geq 1 - \xi_i \\ \xi_i \geq 0 \end{cases}$$

Here, the  $K$  in the constrained formulation and  $C$  in the penalized formulation above have an inverse one-to-one relationship.  $C$  is often called the 'box' constraint because in the Lagrange dual problem, it sets box restriction on the Lagrange dual variable  $\alpha_i$ :

$$0 \leq \alpha_i \leq C$$

## Linear SVM

Looking closely at the problem below,

$$\min_{\beta, \beta_0, (\xi_i)} \frac{1}{2} \|\beta\|^2 + C \sum_{i=1}^n \xi_i \quad \text{s.t.} \quad \begin{cases} \xi_i \geq 1 - y_i(x'_i \beta + \beta_0) \\ \xi_i \geq 0 \end{cases}$$

We can trivially solve the above problem first w.r.t.  $(\xi_i)$  to get an unconstrained problem:

$$\min_{\beta, \beta_0} \frac{1}{2} \|\beta\|^2 + C \sum_{i=1}^n [1 - y_i(x'_i \beta + \beta_0)]_+$$

where  $[1 - y_i(x'_i \beta + \beta_0)]_+ = \max\{0, 1 - y_i(x'_i \beta + \beta_0)\}$  is called hinge loss function.



## Linear SVM

$$\min_{\beta, \beta_0} \frac{1}{2} \|\beta\|^2 + C \sum_{i=1}^n [1 - y_i(x'_i \beta + \beta_0)]_+$$

Here,  $C$  controls the degree to which the second term is more important than the first term in the minimization problem. We may instead write:

$$\min_{\beta, \beta_0} \sum_{i=1}^n [1 - y_i(x'_i \beta + \beta_0)]_+ + \frac{1}{2C} \|\beta\|^2$$

This is the minimization of the hinge loss function with  $l^2$  penalty, where the tuning parameter is given by  $\frac{1}{2C}$ . We see that the box constraint parameter  $C$  in SVM controls the size of the penalty given to the  $l_2$  norm of  $\beta$ .

## Logit model

Suppose,

$$x_i \in \mathbb{R}^p, y_i \in \{-1, 1\}$$

The logit model characterizes the log-likelihood ratio as the linear combination:

$$\log \frac{\mathbb{P}\{Y = 1|X = x\}}{\mathbb{P}\{Y = -1|X = x\}} = \beta_0 + \beta'x$$

This implies that the conditional probability is given by

$$\mathbb{P}\{Y = 1|X = x\} = \frac{1}{1 + e^{-(\beta_0 + \beta'x)}}$$

## Logit model

This model is typically estimated by minimizing the negative log-likelihood.

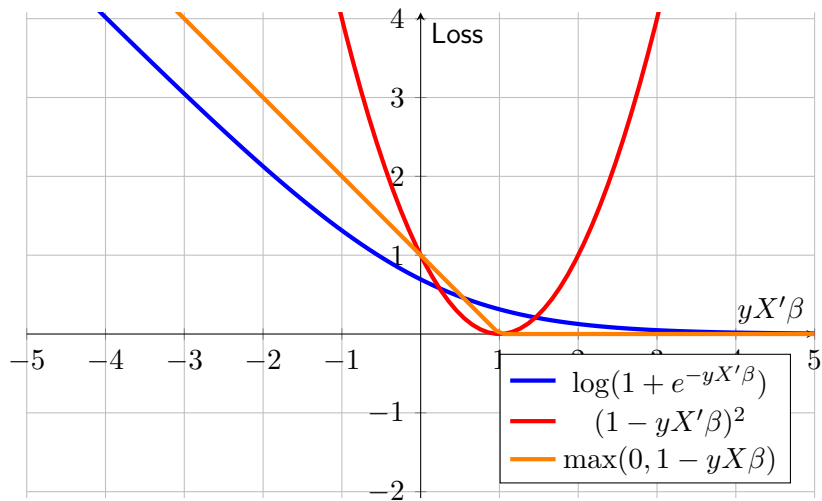
$$\min_{\beta, \beta_0} \sum_{i=1}^n \log(1 + e^{-y_i(\beta_0 + \beta' x_i)})$$

As in ridge regression, we may impose  $l_2$  norm penalty to regularize the variance of  $\hat{\beta}$ .

$$\min_{\beta, \beta_0} \sum_{i=1}^n \log(1 + e^{-y_i(\beta_0 + \beta' x_i)}) + \lambda \|\beta\|^2$$

where  $\lambda$  is the tuning parameter. This can be interpreted as fitting the data by minimizing the binomial deviance loss with  $l_2$  penalty.

# Loss



# SVM vs Logit model

## SVM

$$\min_{\beta, \beta_0} \sum_{i=1}^n [1 - y_i(x'_i \beta + \beta_0)]_+ + \frac{1}{2C} \|\beta\|^2$$

## Logit

$$\min_{\beta, \beta_0} \sum_{i=1}^n \log(1 + e^{-y_i(x'_i \beta + \beta_0)}) + \lambda \|\beta\|^2$$

Using a logit model with  $l_2$  penalty produces very similar results as the ones from linear SVM.

- ▶ Penalized logit model can handle multiple classes and provides class probabilities without Platt-scaling.
- ▶ The solution to SVM is sparse with respect to the samples and thus has a more efficient optimization algorithm.

## Lagrange Duality

## Lagrange Duality

Consider a general minimization problem:

$$\begin{aligned} \min_x \quad & f(x) \quad \text{s.t.} \quad h_i(x) \leq 0, \quad i = 1, \dots, m \\ & l_j(x) = 0, \quad i = 1, \dots, r \end{aligned}$$

We define the **Lagrangian** as

$$L(x, u, v) = f(x) + \sum_{i=1}^m u_i h_i(x) + \sum_{j=1}^r v_j l_j(x)$$

where  $(u_i)_{i=1}^m$  and  $(v_j)_{j=1}^r$  are the Lagrangian multipliers. It is easy to see that as long as  $x$  is feasible and  $u_i \geq 0, i = 1, \dots, m$ , we have

$$f(x) \geq L(x, u, v) \tag{1}$$

## Lagrange Duality

Let  $C$  denote the feasible set. Then, the minimization problem may be written as

$$\min_{x \in C} f(x)$$

We call this problem the **primal problem**. Due to (1), we have

$$\min_{x \in C} f(x) \geq \min_{x \in C} L(x, u, v)$$

as long as  $u \geq 0$ . Minimizing  $L(x, u, v)$  over all  $x$  gives a lower bound to the primal problem.

$$\min_{x \in C} f(x) \geq \min_{x \in C} L(x, u, v) \geq \min_x L(x, u, v) = g(u, v)$$

We call  $g : \mathbb{R}^{m+r} \rightarrow \mathbb{R}$  the **Lagrange dual function**. For any  $u \geq 0$ ,  $g(u, v)$  provides a lower bound to the primal optimal value  $f^*$ .



# Lagrange Duality

To obtain the sharpest lower bound, we solve the **dual problem**

$$\max_{u,v} g(u, v) \quad \text{s.t.} \quad u \geq 0$$

We call the following relation the weak duality:

$$\min_{x \in C} f(x) = f^* \geq g^* = \max_{u \geq 0, v} g(u, v)$$

This always holds, even for non-convex problems. When does the inequality become equality?

## Lagrange Duality

Note that the dual problem is always a convex optimization problem. To see this, it suffices to show that the dual function

$$\begin{aligned} g(u, v) &= \min_x L(x, u, v) \\ &= \min_x \left\{ f(x) + \sum_{i=1}^m u_i h_i(x) + \sum_{j=1}^r v_j l_j(x) \right\} \end{aligned}$$

is concave in  $u$ , and  $v$ . To see this, note that for a fixed  $x$ ,  $L(x, u, v)$  is an affine function in  $u$  and  $v$ , and the pointwise minimum of a set of affine functions is always concave.

**FACT:** The pointwise minimum of a set of affine functions is always concave.

## Strong duality (Slater's condition)

Recall that we always have the weak duality, i.e.,  $f^* \geq g^*$ . Furthermore, if the primal problem is a convex problem, and there exists at least one strictly feasible  $x \in \mathbb{R}^p$ , i.e.,

$$h_i(x) < 0, \quad i = 1, \dots, m$$

and

$$l_j(x) = 0, \quad j = 1, \dots, r$$

Then, the strong duality holds so that we have

$$\begin{aligned} \min_{x \in C} f(x) = f^* & \stackrel{\text{(Strong Duality)}}{=} g^* = \max_{u \geq 0, v} g(u, v) \\ & = \max_{u \geq 0, v} \left( \min_x L(x, u, v) \right) \end{aligned}$$

# Solving the dual of Optimal Separating Hyperplane

Primal Problem:

$$\min_{\beta, \beta_0} \frac{1}{2} \|\beta\|^2 \quad \text{s.t.} \quad y_i(x'_i\beta + \beta_0) \geq 1, \quad i = 1, \dots, n$$

Lagrangian:

$$L(\beta, \beta_0, \alpha) = \frac{1}{2} \|\beta\|^2 + \sum_{i=1}^n \alpha_i [1 - y_i(x'_i\beta + \beta_0)]$$

Lagrange dual function:

$$g(\alpha) = \min_{\beta, \beta_0} L(\beta, \beta_0, \alpha)$$

Dual Problem:

$$\max_{\alpha \geq 0} g(\alpha) \quad \text{s.t.} \quad \alpha_i \geq 0, \quad i = 1, \dots, n$$

## Solving the dual of Optimal Separating Hyperplane

Since the Lagrangian is quadratic in terms of  $\beta$ , and linear in terms of  $\beta_0$ , we can easily obtain dual function:

$$g(\alpha) = \min_{\beta, \beta_0} L(\beta, \beta_0, \alpha) = \min_{\beta, \beta_0} \left( \frac{1}{2} \|\beta\|^2 + \sum_{i=1}^n \alpha_i [1 - y_i(x_i' \beta + \beta_0)] \right)$$

FOC:

$$\beta = \sum_{i=1}^n \alpha_i y_i x_i$$

$$0 = \sum_{i=1}^n \alpha_i y_i$$

Substituting and rearranging yields:

$$g(\alpha) = \begin{cases} \sum_{i=1}^n \alpha_i - \frac{1}{2} \sum_{i=1}^n \sum_{k=1}^n \alpha_i \alpha_k y_i y_k x_i' x_k & \text{if } 0 = \sum_{i=1}^n \alpha_i y_i \\ -\infty & \text{otherwise} \end{cases}$$

## Solving the dual of Optimal Separating Hyperplane

Since the objective function is convex, and there exist strictly feasible points in inequality constraints (Slater's condition), strong duality holds:

$$f^* = \max_{\alpha \geq 0} g(\alpha)$$

and we can get the solution to the original problem by solving the dual:

$$\max_{\alpha \geq 0} g(\alpha) = \max_{\alpha \geq 0} \left( \sum_{i=1}^n \alpha_i - \frac{1}{2} \sum_{i=1}^n \sum_{k=1}^n \alpha_i \alpha_k y_i y_k x_i' x_k \right) \quad \text{s.t.} \quad \sum_{i=1}^n \alpha_i y_i = 0$$

**Question:**

After obtaining  $\alpha_i$ , how can we get  $\beta$  and  $\beta_0$ ?

## Solving the dual of SVM

Let's consider the linear SVM problem:

$$\min_{\beta, \beta_0, (\xi_i)} \frac{1}{2} \|\beta\|^2 + C \sum_{i=1}^n \xi_i \quad \text{s.t.} \quad \begin{cases} \xi_i \geq 1 - y_i(x'_i \beta + \beta_0) \\ \xi_i \geq 0, \quad i = 1, \dots, n \end{cases}$$

The Lagrange function is:

$$L(\beta, \beta_0, \xi_i, \alpha_i, \mu_i) = \frac{1}{2} \|\beta\|^2 + C \sum_{i=1}^n \xi_i + \sum_{i=1}^n \alpha_i [(1 - \xi_i) - y_i(x'_i \beta + \beta_0)] + \sum_{i=1}^n \mu_i (-\xi_i)$$

We can also solve the SVM problem via the Lagrange dual problem. We expect this to be easier because  $\beta, \beta_0, (\xi_i)$  are just quadratic or linear in  $L$ .

## Solving the dual of SVM

The Lagrangian:

$$L(\beta, \beta_0, \xi_i, \alpha_i, \mu_i) = \frac{1}{2} \|\beta\|^2 + C \sum_{i=1}^n \xi_i + \sum_{i=1}^n \alpha_i [(1 - \xi_i) - y_i(x_i' \beta + \beta_0)] + \sum_{i=1}^n \mu_i (-\xi_i)$$

Now we minimize w.r.t the primal variables  $\beta, \beta_0, (\xi_i)$  to get the Lagrange dual function  $g$ .

The FOCs are:

$$\beta) \quad \beta = \sum_{i=1}^n \alpha_i y_i x_i$$

$$\beta_0) \quad 0 = \sum_{i=1}^n \alpha_i y_i$$

$$\xi_i) \quad \alpha_i = C - \mu_i, i = 1, \dots, n$$

We plug this into the Lagrangian.



## Solving the dual of SVM

Plugging in the FOCs, we get<sup>1</sup>:

$$g(\alpha, \mu) = \begin{cases} \sum_{i=1}^n \alpha_i - \frac{1}{2} \sum_{i=1}^n \sum_{k=1}^n \alpha_i \alpha_k y_i y_k x_i' x_k & \text{if } \begin{cases} 0 = \sum_{i=1}^n \alpha_i y_i, \\ \alpha_i = C - \mu_i, i = 1, \dots, n \end{cases} \\ -\infty & \text{Otherwise} \end{cases}$$

Since the dual variables  $\alpha, \mu$  should be greater or equal to 0,  $\alpha_i = C - \mu_i$  can be rewritten as  $0 \leq \alpha_i \leq C$ , neatly eliminating the variable  $\mu$ .<sup>2</sup>

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<sup>1</sup>Note that we get  $g(\alpha, \mu) = -\infty$  when the FOCs from  $\beta_0$  and  $\xi$  are not satisfied. This is because the objective function is linear in these variables.

<sup>2</sup>Here, we see where the term "box" constraint comes from.

## Solving the dual of SVM

The problem reduces to:

$$\max_{(\alpha_i)} g(\alpha) = \max_{(\alpha_i)} \left( \sum_{i=1}^n \alpha_i - \frac{1}{2} \sum_{i=1}^n \sum_{k=1}^n \alpha_i \alpha_k y_i y_k x_i' x_k \right) \text{ s.t. } \begin{cases} 0 = \sum_{i=1}^n \alpha_i y_i \\ 0 \leq \alpha_i \leq C \end{cases}$$

Note that  $\alpha_i$ 's are Lagrange multipliers associated with the constraint:

$$y_i(x_i' \beta + \beta_0) \geq 1 - \xi_i.$$

**Question:**

After obtaining  $\alpha_i$ , how can we get  $\beta, \beta_0, (\xi_i)$ ?

## Karush-Kuhn-Tucker conditions

# Karush-Kuhn-Tucker conditions

Given general problem

$$\begin{array}{ll}\min_x & f(x) \\ \text{s.t.} & h_i(x) \leq 0, \quad i = 1, \dots, m \\ & l_j(x) = 0, \quad i = 1, \dots, r\end{array}$$

The **Karush-Kuhn-Tucker conditions** are:

- ▶ (stationarity):  $0 \in \partial \left( f(x) + \sum_{i=1}^m u_i h_i + \sum_{j=1}^r v_j l_j(x) \right)$
- ▶ (complementary slackness):  $u_i \cdot h_i(x) = 0$  for all  $i$
- ▶ (primal and dual feasibility):  $h_i(x) \leq 0, l_j(x) = 0$  for all  $i, j$  and  $u_i \geq 0$  for all  $i$ .

## Necessity

### Claim:

Let  $x^*$  and  $u^*, v^*$  be primal and dual solutions with zero duality gap. Then, KKT conditions are satisfied.

### Proof:

$$\begin{aligned} f(x^*) &= g(u^*, v^*) \\ &= \min_x \left( f(x) + \sum_{i=1}^m u_i^* h_i(x) + \sum_{j=1}^r v_j^* l_j(x) \right) \\ &\leq f(x^*) + \sum_{i=1}^m u_i^* h_i(x^*) + \sum_{j=1}^r v_j^* l_j(x^*) \\ &\leq f(x^*) \end{aligned}$$

Here, the first equality uses the zero duality gap, and the second equality comes from the definition of we get Lagrange dual function. The first inequality comes from minimization, and the second inequality comes from the primal and dual feasibility.

## Necessity

This means that

$$\begin{aligned}\min_x \left( f(x) + \sum_{i=1}^m u_i^* h_i(x) + \sum_{j=1}^r v_j^* l_j(x) \right) &= f(x^*) + \sum_{i=1}^m u_i^* h_i(x^*) + \sum_{j=1}^r v_j^* l_j(x^*) \\ &= f(x^*)\end{aligned}$$

We obtain the stationarity condition and complementary slackness condition from the first and second equality respectively. □

## Sufficiency

### Claim:

If  $x^*, u^*, v^*$  satisfy the KKT conditions, then they are solutions to the primal and dual problems.

### Proof

We have

$$\begin{aligned} g(u^*, v^*) &= \min_x \left( f(x) + \sum_{i=1}^m u_i^* h_i(x) + \sum_{j=1}^r v_j^* l_j(x) \right) \\ &= f(x^*) + \sum_{i=1}^m u_i^* h_i(x^*) + \sum_{j=1}^r v_j^* l_j(x^*) \\ &= f(x^*) \end{aligned}$$

where the first equality comes from the definition, the second equality comes from the stationarity condition, and the final equality comes from complementary slackness. Since the duality gap is zero,  $x^*, u^*, v^*$  are the solutions to the primal and dual problems.

# Optimal Separating Hyperplane

Primal Problem:

$$\min_{\beta, \beta_0} \frac{1}{2} \|\beta\|^2 \quad \text{s.t.} \quad y_i(x'_i \beta + \beta_0) \geq 1, \quad i = 1, \dots, n$$

Lagrangian:

$$L(\beta, \beta_0, \alpha) = \frac{1}{2} \|\beta\|^2 + \sum_{i=1}^n \alpha_i [1 - y_i(x'_i \beta + \beta_0)]$$

Dual Problem:

$$\max_{\alpha \geq 0} g(\alpha) \left\{ = \sum_{i=1}^n \alpha_i - \frac{1}{2} \sum_{i=1}^n \sum_{k=1}^n \alpha_i \alpha_k y_i y_k x'_i x_k \right\} \quad \text{s.t.} \quad \sum_{i=1}^n \alpha_i y_i = 0$$



## Optimal Separating Hyperplane

Since the strong duality holds for this case, KKT conditions states that if  $\beta, \beta_0$  are optimal they must satisfy the following:

- (1) Stationarity:  $\beta = \sum_{i=1}^n \alpha_i y_i x_i, \quad 0 = \sum_{i=1}^n \alpha_i y_i$
- (2) Complementary slackness:  $\alpha_i [y_i(x'_i \beta + \beta_0) - 1] = 0, \quad i = 1, \dots, n$
- (3) Primal and dual feasibility:  $y_i(x'_i \beta + \beta_0) - 1 \geq 0, \quad \alpha_i \geq 0$

Implications:

- By (1),  $\beta$  is defined only in terms of samples with  $\alpha_i > 0$ . Such samples are called support vectors.
- By (2), if the  $i$ -th sample is support vector (if  $\alpha_i > 0$ ), then  $y_i(x'_i \beta + \beta_0) = 1$  (the sample is on the margin). Also, if  $y_i(x'_i \beta + \beta_0) > 1$  ( $i$ -th sample is not on the margin), then  $\alpha_i = 0$  (the sample is not a support vector).

Primal Problem:

$$\min_{\beta, \beta_0, (\xi_i)} \frac{1}{2} \|\beta\|^2 + C \sum_{i=1}^n \xi_i \quad \text{s.t.} \quad \begin{cases} \xi_i \geq 1 - y_i(x'_i \beta + \beta_0) \\ \xi_i \geq 0, \quad i = 1, \dots, n \end{cases}$$

The Lagrange dual function:

$$g(\alpha, \mu) = \begin{cases} \sum_{i=1}^n \alpha_i - \frac{1}{2} \sum_{i=1}^n \sum_{k=1}^n \alpha_i \alpha_k y_i y_k x'_i x_k & \text{if } \begin{cases} 0 = \sum_{i=1}^n \alpha_i y_i, \\ \alpha_i = C - \mu_i, i = 1, \dots, n \end{cases} \\ -\infty & \text{Otherwise} \end{cases}$$

Dual Problem:

$$\max_{(\alpha_i)} g(\alpha) \{ = \sum_{i=1}^n \alpha_i - \frac{1}{2} \sum_{i=1}^n \sum_{k=1}^n \alpha_i \alpha_k y_i y_k x'_i x_k \} \quad \text{s.t.} \quad \begin{cases} 0 = \sum_{i=1}^n \alpha_i y_i \\ 0 \leq \alpha_i \leq C \quad \forall i = 1, \dots, n \end{cases}$$

## SVM

Since the strict duality also holds in this case,

$$f^* = \max_{\alpha_i, \mu_i} g(\alpha) \left\{ = \sum_{i=1}^n \alpha_i - \frac{1}{2} \sum_{i=1}^n \sum_{k=1}^n \alpha_i \alpha_k y_i y_k x_i' x_k \right\}$$

under the constraint:  $0 = \sum_{i=1}^n \alpha_i y_i$  and  $0 \leq \alpha_i \leq C$ . KKT conditions include:

(1) stationarity:

$$\beta = \sum_{i=1}^n \alpha_i y_i x_i, \quad 0 = \sum_{i=1}^n \alpha_i y_i, \quad \alpha_i = C - \mu_i, \forall i$$

(2) complementary slackness:

$$\alpha_i [y_i (x_i' \beta + \beta_0) - (1 - \xi_i)] = 0$$
$$\mu_i \xi_i = 0 \quad \forall i$$

(3) primal and dual feasibility:

$$y_i (x_i' \beta + \beta_0) - (1 - \xi_i) \geq 0, \quad \xi_i \geq 0, \quad \alpha_i \geq 0, \quad \mu_i \geq 0$$

At optimality, we have  $\beta = \sum_{i=1}^n \alpha_i y_i x_i$ . As before, samples with  $\alpha_i > 0$  is called support vectors. By complementary slackness condition, this can only happen if  $y_i(x'_i\beta + \beta_0) = (1 - \xi_i)$ .

Looking at KKT conditions:

1. If  $\alpha_i = 0$ , then  $\mu_i = C$ ,  $\xi_i = 0$ , and  $y_i(x'_i\beta + \beta_0) - 1 \geq 0$ .
2. If  $0 < \alpha_i < C$ , then  $0 < \mu_i < C$ ,  $\xi_i = 0$ , and  $y_i(x'_i\beta + \beta_0) = 1$ .
3. If  $\alpha_i = C$ , then  $\mu_i = 0$ ,  $\xi_i \geq 0$ , and  $y_i(x'_i\beta + \beta_0) = (1 - \xi_i)$ :  $y_i(x'_i\beta + \beta_0) \leq 1$

We see that samples with  $y_i(x'_i\beta + \beta_0) > 1$  are not a support vector. Also, if a sample is a support vector, the sample is either on the edge of the margin  $y_i(x'_i\beta + \beta_0) = 1$  or inside the margin  $y_i(x'_i\beta + \beta_0) \leq 1$ .